



Just-in-time

The system of production in which an actual order is the trigger for an item to be manufactured is called just-in-time. It enables a firm to produce only the items required, in the right amount, at the right time.

How it works

Also called demand-pull production, just-in-time means that stock levels of raw materials, components, work in progress, and finished goods are kept as low as possible, reducing costs. The system requires detailed planning, scheduling, and flow of resources throughout the production process, now assisted by sophisticated production-scheduling software. Supplies have to be delivered directly to the production line when they are needed, requiring strong relationships and interconnected systems with suppliers. The benefits of reduced inventory are balanced against the cost

of frequent deliveries and loss of purchasing economies of scale (discounts for bulk buying). The system dates back to 1953, the year Toyota brought in just-in-time manufacturing. The phrase is sometimes used in a more general sense today to mean eliminating waste of resources.

JUST-IN-TIME PROS AND CONS

Pros

- **Lower stock** so less storage space and less working capital needed
- **Demand-pull** avoids obsolete, out-of-date stock
- **Staff** spend less time checking and moving items

Cons

- **No room for error**—for instance, if there are any faults in the stock delivered, the whole day's production is halted
- **Operation** is reliant on suppliers
- **No cushion** for sudden upsurge in demand

